

Music Genre Classification: A Replication Study

Alessandro Potenza Camilla Sed

Politecnico di Milano

Scuola di Ingegneria Industriale e dell'Informazione

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Introduction: The Task

What is Music Genre Classification (MGC)?

MGC is a fundamental task in Music Information Retrieval (MIR) that automatically categorizes music tracks into predefined genres based on their audio content.

Applications:

- Music Recommendation
- Library Organization
- Content Discovery

The Reference Paper:

- "Novel mathematical model..." by Patil et al. [3].
- Proposed a U-Net-like architecture.
- **Claimed a remarkable 99.41% accuracy** on the GTZAN dataset.

Our Project Goals

1. **Replicate and Validate:** Implement the U-Net architecture and rigorously evaluate its performance on GTZAN to validate the paper's central claim.
2. **Compare and Analyze:** Compare this architecture against two other common CNN designs to understand its relative strengths.
3. **Extend and Generalize:** Assess the model's robustness by testing it on three additional, diverse datasets.

The Datasets

A diverse set of datasets to test replication, complexity, and generalization.

- **GTZAN**: The primary dataset for our replication study. 1,000 tracks across 10 genres. The standard benchmark.
- **FMA Small**: A more challenging and realistic benchmark. 8,000 tracks across 8 genres.
- **Indian Music Genre (IMG)**: A non-Western dataset to test domain adaptation. 5 genres.
- **Tabla Taala**: A specialized rhythmic dataset to test fine-grained pattern recognition. 8 rhythmic cycles.

Feature Extraction: From Audio to Image

Why Mel-Spectrograms?

! We convert 1D audio signals into 2D image-like representations to leverage the power of CNNs. Mel-spectrograms are perceptually motivated and align with human hearing, making them an ideal feature for audio tasks.

This process involves:

1. Loading and Framing the audio signal.
2. Applying a Short-Time Fourier Transform (STFT).
3. Converting to the Mel scale.
4. Applying Logarithmic Scaling (decibels).

Visualizing the Features: Waveform vs. Mel-Spectrogram

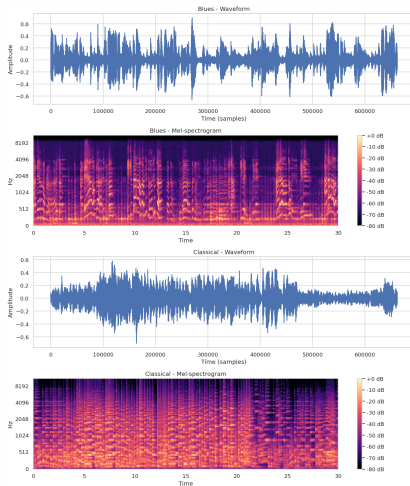


Figure: Comparison for Blues (top) and Classical (bottom) tracks. The CNN learns to recognize these distinct visual textures.

Data Partitioning and Slicing

Slicing Strategy

| Each 30-second track was sliced into 10 non-overlapping 3-second segments. This increases the training data tenfold (e.g., 600 training tracks → 6,000 samples).

Crucial Step: Avoiding Data Leakage

| The dataset was split into train/validation/test sets (60/20/20) **at the track level, BEFORE slicing**. This is vital to prevent the model from "cheating" by recognizing a song's unique fingerprint across different sets.

Model Architectures: The Competitors

We implemented three distinct CNNs for a comprehensive comparison.

1. Efficient-VGG

- Lightweight baseline.
- Simple VGG structure + SE blocks.
- **35k parameters.**

2. ResSE-AudioCNN

- Deeper, more powerful.
- Based on Residual Networks (ResNets) + SE blocks.
- **1.23M parameters.**

Model Architectures: The Champion (U-Net Classifier)

Our Implementation of the Patil et al. Model

| A U-Net-based architecture adapted for classification.

- It uses only the **encoder (contracting) path** of a standard U-Net to learn hierarchical features at multiple resolutions.
- Three down-sampling blocks followed by a bottleneck.
- A **Global Average Pooling layer** and a Dense layer perform the classification.
- **1.18M parameters.**

Training Protocol

For a fair and reproducible comparison, all models were trained with a standardized protocol.

- **Optimizer:** Adam (initial $lr = 1e-3$).
- **Loss:** Categorical Cross-Entropy.
- **Epochs/Batch:** Up to 120 epochs, batch size of 48.
- **Callbacks:** EarlyStopping, ReduceLROnPlateau, ModelCheckpoint.
- **Reproducibility:** All random seeds were fixed to 42.

Let's see the key hyperparameters...

Key Training Hyperparameters

Table: Key training hyperparameters used across experiments.

Hyperparameter	Value
Optimizer	Adam (initial lr = 1×10^{-3})
Batch size (single-split)	48
Batch size (5-fold CV)	16
Epochs (max)	120 (single-split), 60 (CV)
EarlyStopping patience	25 (single-split), 10 (CV)
ReduceLROnPlateau factor	0.2
Mixed precision	enabled (mixed_float16)
Random seeds	Python/NumPy/TensorFlow = 42

GTZAN Results: Performance & Efficiency

U-Net Audio Classifier is the Winner

It achieves the best balance of accuracy and computational efficiency among the tested models.

Here's a summary of the performance on the GTZAN test set.

Performance and Efficiency Metrics (GTZAN Test Set)

Table: Performance and efficiency metrics on the GTZAN test set.

Model	Test Accuracy	Parameters	Latency (ms)
UNet_Audio_Classifier	0.823	1.18 M	23.07
ResSE_Audio_CNN	0.798	1.23 M	26.75
Efficient_VGG	0.752	0.03 M	14.77

Accuracy vs. Efficiency Trade-off

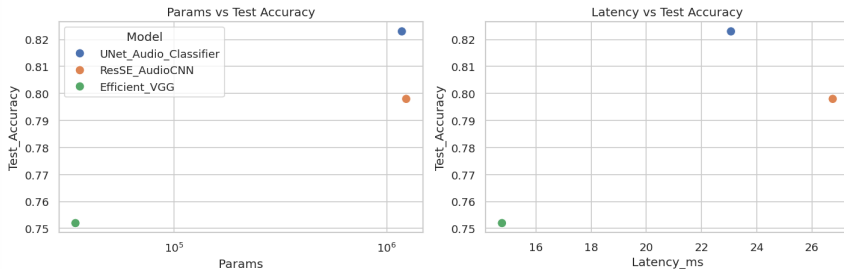


Figure: The U-Net model (blue) offers the best accuracy-efficiency trade-off compared to the comparable ResNet-based model (orange).

Ablation Study: The Impact of SpecAugment

What is SpecAugment?

| A data augmentation technique that randomly masks frequency bands and time steps in the spectrogram, forcing the model to learn more robust features.

Question: Does it always help? **Finding:** Only the U-Net architecture, our largest model, benefited. This suggests model capacity is crucial for leveraging this type of regularization. Simpler models may treat it as noise.

Impact of SpecAugment on Test Accuracy

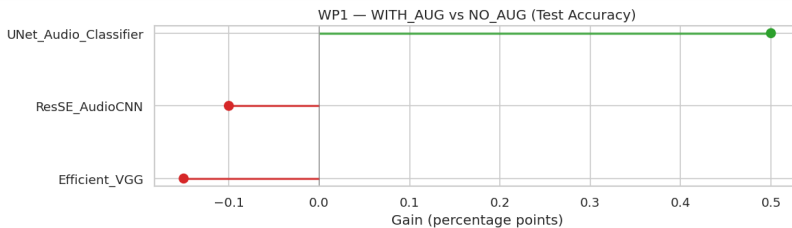


Figure: Change in test accuracy. Only the U-Net shows a positive gain.

Robustness Check: 5-Fold Cross-Validation

A single train/test split can be misleading. We performed a 5-fold cross-validation on the champion U-Net model for a more reliable performance estimate.

Our Most Reliable Result on GTZAN

| 90.44% Mean Validation Accuracy

This result is substantially higher than our single-split accuracy (82.3%) and has a very low standard deviation (0.68%), demonstrating the model's stability.

Analysis of the Winner Model's Performance

Where does the model succeed and fail?

- The model excels on genres with distinct acoustic signatures.
- Its primary weakness lies with genres that are spectrally similar.

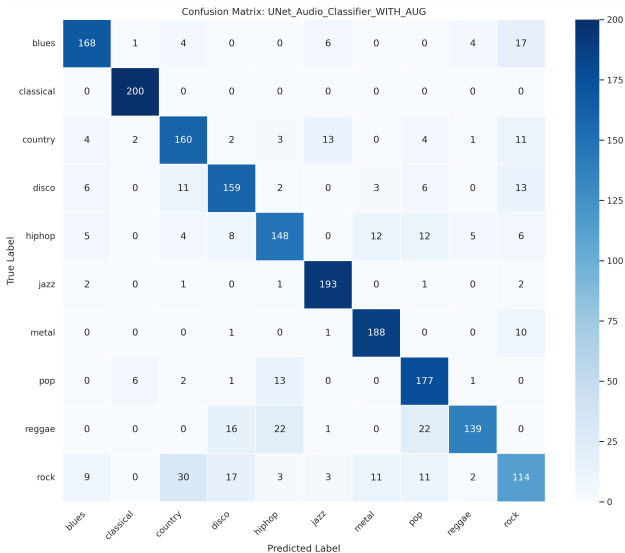
Let's look at the per-class metrics.

Per-Class Classification Report (U-Net on GTZAN)

Table: Per-class classification report for the U-Net model on GTZAN.

Genre	Precision	Recall	F1-Score	Support
blues	0.87	0.84	0.85	200
classical	0.96	1.00	0.98	200
country	0.75	0.80	0.78	200
disco	0.78	0.80	0.79	200
hiphop	0.77	0.74	0.76	200
jazz	0.89	0.96	0.93	200
metal	0.88	0.94	0.91	200
pop	0.76	0.89	0.82	200
reggae	0.91	0.69	0.79	200
rock	0.66	0.57	0.61	200
Accuracy			0.82	2000
Macro Avg	0.82	0.82	0.82	2000
Weighted Avg	0.82	0.82	0.82	2000

GTZAN Confusion Matrix: Visualizing the Errors



Analysis of the Confusion Matrix

The confusion matrix confirms our findings:

- A strong diagonal indicates high overall accuracy.
- The low recall for 'rock' is explained by its confusion with spectrally similar genres like 'country' and 'disco'.
- Similarly, 'reggae' is often confused with 'hiphop'.
- These errors are musically coherent, highlighting the intrinsic difficulty of the task based on short 3-second clips.

Beyond GTZAN: Cross-Dataset Generalization

We tested our champion U-Net model on three new datasets to assess its true capabilities.

Strategy 1: From Scratch

- **Dataset:** FMA Small
- **Challenge:** High complexity and realism.

Strategy 2: Transfer Learning

- **Datasets:** Indian Music & Tabla Taala
- **Challenge:** Domain shift (cultural and task-specific).

Performance Across All Datasets

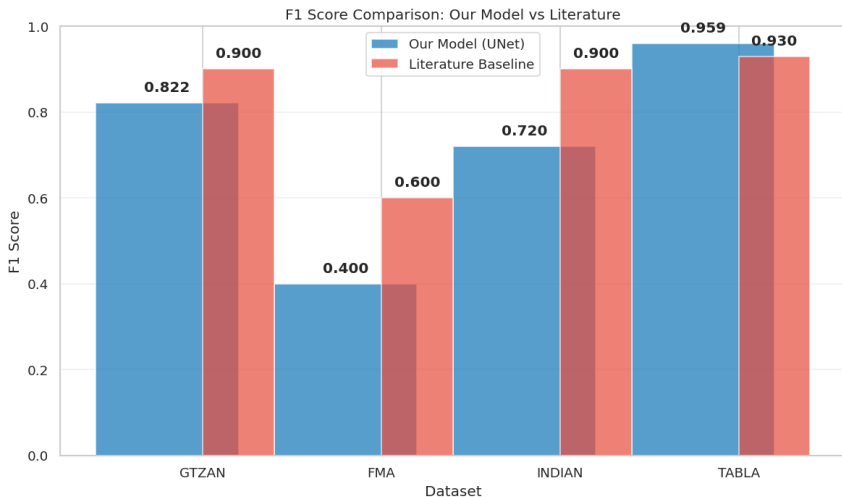


Figure: The success of transfer learning on Indian and Tabla datasets is evident

Generalization Results: A Tale of Two Strategies

Transfer Learning is Highly Effective

Features learned on Western music (GTZAN) can be successfully adapted to new domains.

- **FMA Small (from scratch):** 41.1% accuracy. Much better than random (12.5%), but shows the dataset's difficulty.
- **Indian Music (transfer):** 72.2% accuracy. Strong result, confirming feature universality.
- **Tabla Taala (transfer):** **96.5% accuracy.** Exceptional result, showing the model learned to recognize fine-grained rhythmic patterns.

Visualizing Generalization Performance

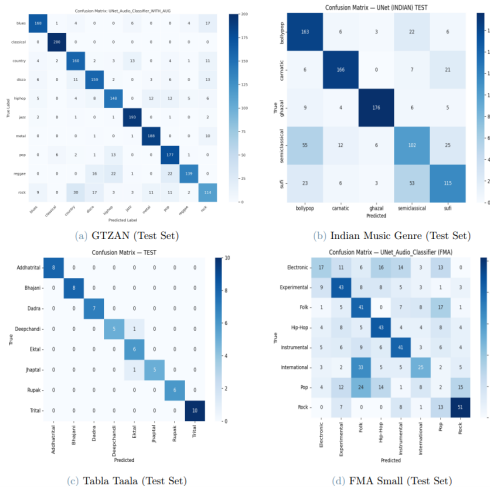


Figure: Strong diagonals for GTZAN and Tabla Taala vs. a diffuse matrix for FMA Small, reflecting dataset complexity.

Discussion: The Big Question

Why is there such a large gap between our results and the reference paper?

Patil et al. [3]

99.41%

Our Best Result (5-Fold
CV)

90.44%

A difference of **~ 9 percentage points** is too significant to be caused by minor architectural variations.

Potential Sources of Discrepancy

Our analysis points to methodological factors, not architectural ones.

1. **Data Splitting and Information Leakage (Leading Hypothesis):**

- The original paper does not specify its splitting protocol.
- Randomly splitting *after* slicing audio tracks is a common pitfall that transforms the task into *audio fingerprinting*, leading to inflated scores.
- Our strict track-level split provides a more realistic evaluation.

2. **Methodological Ambiguity:** The paper's "novel mathematical model" lacks specific details for perfect replication.

3. **Context of State-of-the-Art on GTZAN:** Accuracies of 90-95% are typical. A near-perfect score of 99.41% is a significant outlier.

Corroboration of Model Efficiency

One thing we can confirm...

Despite the accuracy discrepancy, our efficiency analysis validates a central premise of the paper.

Our experiments confirm that the U-Net based classifier provides a **superior balance of performance and computational cost** compared to a standard ResNet-based architecture.

- **UNet_Audio_Classifier**: 82.3% accuracy, 1.18M params, 23.07 ms latency.
- **ResSE-AudioCNN**: 79.8% accuracy, 1.23M params, 26.75 ms latency.

The U-Net's multi-scale encoder design is an effective choice for this task.

Conclusion & Key Findings

- **U-Net Encoder is Effective:** We corroborate the architectural premise. Our champion model achieved a robust **90.44%** accuracy on GTZAN, establishing a new, transparent, and reproducible benchmark.
- **Performance Gap is Methodological:** The 99.41% claim is an outlier, likely attributable to data leakage from an improper splitting protocol, not a superior architecture.
- **Adaptability Through Transfer Learning:** The architecture shows excellent adaptability, achieving 96.5% accuracy on the specialized Tabla Taala dataset.
- **Importance of Rigor:** This work serves as a case study on the paramount importance of methodological transparency and robust validation in machine learning.